

DAY 01: PHYSICAL SERVER, VIRTUAL SERVER & GIT COMMANDS

Cloud Computing & DevOps Master Curriculum

1. Physical Server vs Virtual Server

A **Physical Server** (Bare Metal) is a single physical computer hardware dedicated entirely to running a single operating system and application workload. It provides maximum performance and direct access to hardware resources, but suffers from high upfront cost, slow provisioning times, and lower resource utilization.

A **Virtual Server** (Virtual Machine / VM) is a software-defined computer that runs on top of a physical server using a **Hypervisor** (e.g., VMware ESXi, KVM, Hyper-V). Virtualization allows partitioning a physical server into multiple isolated VMs, each with its own vCPU, RAM, storage, and operating system.

Physical Server (Bare Metal)	Virtual Server (VM)
Dedicated physical hardware	Shared hardware via hypervisor
Hours to Days	Seconds to Minutes
Manual hardware upgrade	Elastic / Software scalable
Often low (15-30%)	High (up to 80-90%)
High CapEx (Capital Expenses)	Low OpEx (Pay-as-you-go)

2. Essential Git Commands

Git is a distributed version control system (DVCS) that tracks changes in source code during software development.

Initialize a new Git repository:

```
git init
```

Clone an existing repository:

```
git clone https://github.com/username/repository.git
```

Check status of staged and unstaged files:

```
git status
```

Stage changes for commit:

```
git add index.html
git add .
```

Commit staged changes with message:

```
git commit -m 'Initial commit: Added project homepage'
```

View commit history:

```
git log --oneline --graph --all
```

Push commits to remote repository:

```
git push -u origin main
```

KEY TAKEAWAYS & SUMMARY

- ✓ Virtualization decouples software from physical hardware, enabling cloud elasticity.
- ✓ Hypervisors manage and allocate physical RAM, CPU, and disk to Virtual Machines.
- ✓ Git tracks code changes incrementally across three states: Working Directory, Staging Area, and Commit History.